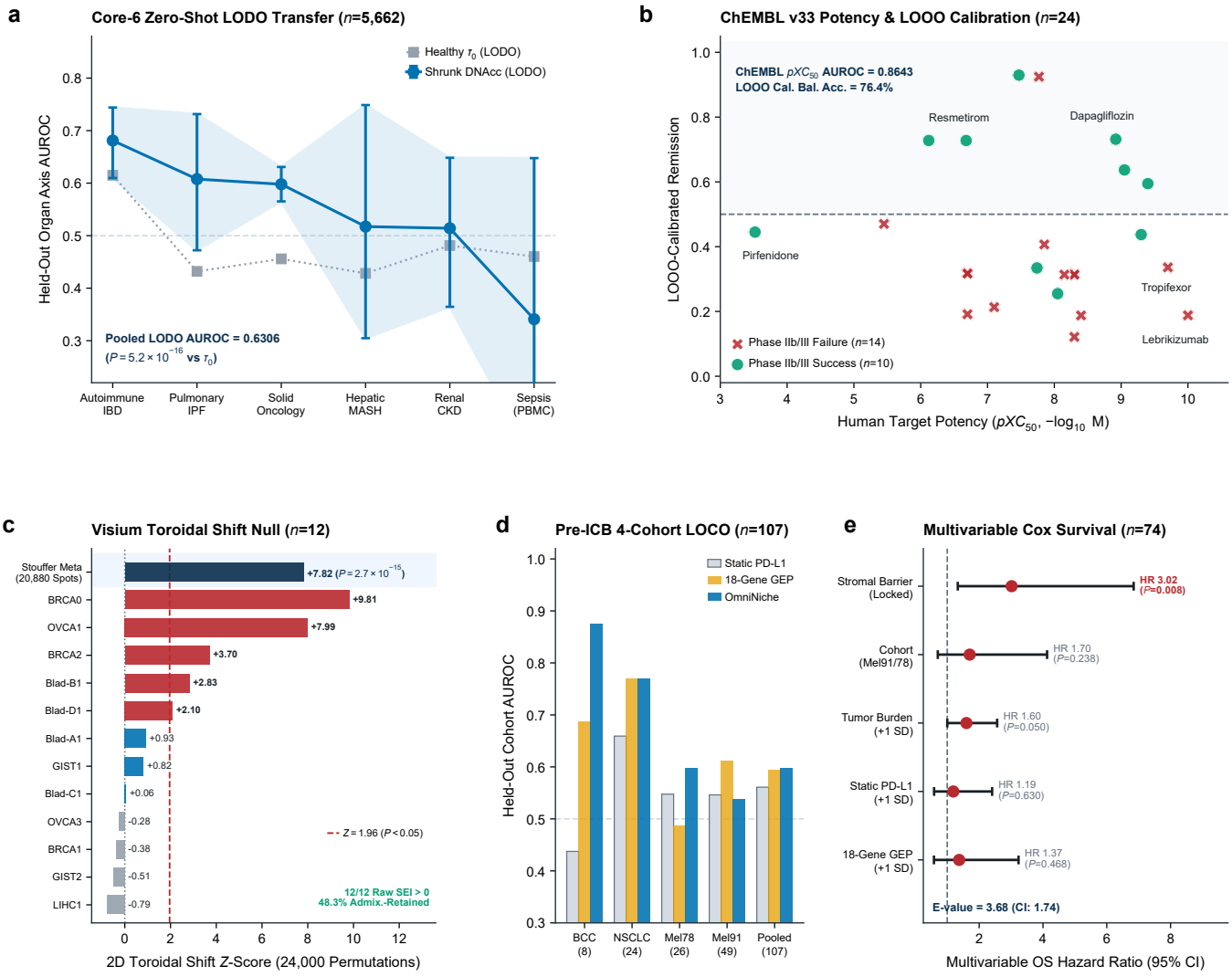

a
Core-6 Zero-Shot LODO Transfer (n=5,662)
Healthy τ0 (LODO)
Shrunk DNAcc (LODO)
Held-Out Organ Axis AUROC
Pooled LODO AUROC = 0.6306
(P = 5.2 × 10−16 vs τ0)
Autoimmune IBD
Pulmonary IPF
Solid Oncology
Hepatic MASH
Renal CKD
Sepsis (PBMC)
b
ChEMBL v33 Potency & LOOO Calibration (n=24)
ChEMBL pXC50 AUROC = 0.8643
LOOO Cal. Bal. Acc. = 76.4%
Resmetirom
Dapagliflozin
Pirfenidone
Tropifexor
Lebrikizumab
Phase IIb/III Failure (n=14)
Phase IIb/III Success (n=10)
LOOO-Calibrated Remission
Human Target Potency (pXC50, −log10 M)
c
Visium Toroidal Shift Null (n=12)
Stouffer Meta (20,880 Spots)
+7.82 (P = 2.7 × 10−15)
BRCA0
+9.81
OVCA1
+7.99
BRCA2
+3.70
Blad-B1
+2.83
Blad-D1
+2.10
Blad-A1
+0.93
GIST1
+0.82
Blad-C1
+0.06
OVCA3
-0.28
BRCA1
-0.38
GIST2
-0.51
LIHC1
-0.79
Z = 1.96 (P < 0.05)
12/12 Raw SEI > 0
48.3% Admix.-Retained
2D Toroidal Shift Z-Score (24,000 Permutations)
d
Pre-ICB 4-Cohort LOCO (n=107)
Static PD-L1
18-Gene GEP
OmniNiche
Held-Out Cohort AUROC
BCC (8)
NSCLC (24)
Mel78 (26)
Mel91 (49)
Pooled (107)
e
Multivariable Cox Survival (n=74)
Stromal Barrier (Locked)
HR 3.02 (P=0.008)
Cohort (Mel91/78)
HR 1.70 (P=0.238)
Tumor Burden (+1 SD)
HR 1.60 (P=0.050)
Static PD-L1 (+1 SD)
HR 1.19 (P=0.630)
18-Gene GEP (+1 SD)
HR 1.37 (P=0.468)
E-value = 3.68 (CI: 1.74)
Multivariable OS Hazard Ratio (95% CI)